Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2024

Time: 03 Hours
Full Marks: 70

Course : B.Tech
Code : 104703

Subject: Wireless Communication

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct answer of the following (any seven question only):* **[2 x 7 = 14]**

(a) Which of the following is a protocol used for cordless telephone system
(i) ERMES (ii) IS-95
(iii) FLEX (iv) PACS

(b) In which frequency range do the cordless phones mostly work?
(i) 540-1600 KHz (ii) 200-540 KHz
(iii) 88-108 MHz (iv) 43-50 MHz

(c) Time division duplexing uses _____ to provide both a forward and reverse link
(i) Frequency (ii) Frequency
(iii) Time and frequency (iv) Cell spacing

(d) Local wireless system is
(i) GSM (ii) Cordless telephone system
(iii) UMTS (iv) EDGE

(e) Frequency division multiple access (FDMA) assigns _____ channels to _____ users
(i) Individual, individual (ii) Many, individual
(iii) Individual, many (iv) Many, many

(f) Polarization diversity uses the _____ as the diversity element.
(i) Modulation index (ii) Carrier frequency
(iii) Reflection coefficient (iv) Coherence time

(g) Who sets the standards of GSM?
(i) ITU (ii) ETSI
(iii) AT & T (iv) USDC

(h) _____ is used to prevent deep fade for rapidly varying channel
(i) Modulation (ii) Demodulation
(iii) Macroscopic diversity technique (iv) Microscopic diversity technique

(i) _____ is undesired RF radiation
(i) Intermodulation frequency (ii) Intermediate frequency
(iii) Instantaneous frequency (iv) Instrumental frequency

(j) Which of the following is a fully digital cordless system?
(i) CT0
(iii) CT1+ (ii) CT1
(iv) DECT

Q.2 (a) What do you understand by Global system for Mobile Communication? Explain GSM Architecture with neat sketch diagram. [7]

(b) Discuss about the BER Performance in Fading Channels. [7]

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Q.3 (a) Discuss the cellular concept and Multiple Access Techniques? Also define the FDMA. [7]

(b) Differentiate between GSM and GPRS. [7]

Q.4 (a) Explain Fast Fading Wireless Channel Modelling. [7]

(b) Explain the Third Generation (3G) Mobile Services. And how it is differ from 2G? [7]

Q.5 (a) Write short note on Wireless LANs (IEEE 802.11). [7]

(b) Explain Rayleigh/Ricean Fading Channels? Compare Rayleigh fading with Fast Fading Wireless Modelling. [7]

Q.6 (a) Compare FDMA, TDMA, and CDMA with suitable diagram. [7]

(b) Explain the Cordless telephone systems in Wireless Mobile communication. [7]

Q.7 (a) Briefly explain the 2.5G Mobile Data Networks? [7]

(b) Explain the GPRS architecture in Mobile Data Network. [7]

Q.8 (a) Explain the concept of Diversity modelling for Wireless Communications. [7]

(b) Define the Wideband code division multiple access in 2.5G Mobile Data Networks. [7]

Q.9 Write short notes on ***any two*** of the following:- [7 x 2 = 14]

(a) Paging Systems

(b) CDMA

(c) 2G Networks

(d) Diversity

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